FUNCTIONS IN C PROGRAMMING

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1. Introduction to Functions in C

A **function** is a block of code that performs a specific task. Functions help break a large program into smaller, manageable parts.

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2. Why We Need Functions in C

Functions are needed in C programming because:

- They divide a program into smaller modules
- They reduce code repetition
- They make programs easy to read and understand
- They simplify debugging and maintenance
- They improve reusability of code

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3. Advantages of Functions

- **Modularity** – Program is divided into parts
- **Reusability** – Same function can be used many times
- **Easy Debugging** – Errors are easy to find
- **Easy Maintenance** – Changes can be made easily
- **Saves Time and Effort**

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4. Types of Functions

1. Library Functions

Predefined functions provided by C
Examples: printf(), scanf(), strlen()

2. User-defined Functions

Functions created by the programmer
Example: add(), factorial()

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5. Parts of a Function

1. **Function Declaration (Prototype)**

```
int add(int, int);
```

2. **Function Definition**

```
int add(int a, int b) {
   return a + b;
}
```

3. **Function Call**

```
sum = add(5, 3);
```

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6. Function Syntax

```
return_type function_name(parameters) {
   // statements
   return value;
}
```

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7. Types of Functions Based on Arguments & Return Value

1. No arguments, no return value
2. Arguments, no return value
3. No arguments, return value
4. Arguments and return value

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8. Example Program (Function with Arguments and Return Value)

```
#include <stdio.h>

int add(int a, int b) {
   return a + b;
}

int main() {
   int sum;
   sum = add(10, 20);
   printf("Sum = %d", sum);
```

```
    return 0;
}
```

Explanation:

- add() function adds two numbers
- main() calls the function
- Result is returned and printed

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9. Main Function

- Execution of a C program starts from main()
- Only one main() function is allowed

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10. Conclusion

Functions make C programs **simple, reusable, organized, and easy to maintain**. They are an important concept in structured programming.